

CPU, ALU, Control Unit & Registers

Lesson 05 — Functional Units and Registers

JKSSB Computer Awareness

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Video Lesson 05

Watch: CPU Explained: ALU, Control Unit & Registers

The CPU and Its Functional Units

- ▶ The **Central Processing Unit (CPU)** is often called the “brain” of the computer.
- ▶ It fetches, decodes, and executes instructions and works with main memory.
- ▶ Its main functional units are the Arithmetic Logic Unit (ALU), the Control Unit (CU), and the registers.
- ▶ Data and control signals move between these units over the internal buses.

Memory hook

ALU computes, the Control Unit coordinates, and the registers hold the working data.

The Three Functional Units at a Glance

ALU

Performs arithmetic and logical operations on binary data.

Control Unit

Directs and sequences operations; issues control signals.

Registers

Smallest, fastest storage, located inside the CPU.

Keep the roles separate; most wrong answers swap the ALU and the Control Unit.

The Arithmetic Logic Unit (ALU)

- ▶ Performs **arithmetic** operations: addition, subtraction, increment, decrement, multiplication, and division.
- ▶ Performs **logical** operations: AND, OR, NOT, XOR, shifts, and comparisons.
- ▶ Operates on **binary data** using logic-gate operations.
- ▶ Returns its result to a register or the accumulator; it does not permanently store data.

Exam cue

The ALU is the part that computes; its operations are carried out with logic gates on binary values.

ALU Works on Binary Data with Logic Gates

- ▶ Inside the CPU, numbers and characters are represented in binary.
- ▶ Logic gates such as AND, OR, NOT, and XOR implement the ALU's operations.
- ▶ Example: adding 1 and 1 works on the binary values, and the result is placed where it can be reused.
- ▶ **ALU** stands for Arithmetic Logic Unit.

Remember

Arithmetic and logic are the two halves of the ALU's name; both run on binary data.

The Control Unit (CU)

- ▶ **Directs and sequences** the operations of the computer.
- ▶ Controls the **flow of data** between the ALU, registers, and memory.
- ▶ Decodes instructions and **issues control signals** that time and trigger the other units.
- ▶ It does **not** perform arithmetic or logical calculations.

Role in one line

The Control Unit is the conductor: it tells every part what to do and when, but it does not itself compute.

Control Unit vs ALU: Do Not Swap the Roles

Unit	Primary job
Arithmetic Logic Unit (ALU)	Performs arithmetic and logic operations
Control Unit (CU)	Directs operations and issues control signals

Trap alert

The Control Unit controls and coordinates; it does **not** add or subtract. Saying “CU computes” or “ALU controls” is wrong.

Registers: Fastest Storage Inside the CPU

- ▶ **Registers** are the **smallest and fastest** storage in a computer.
- ▶ They sit inside the CPU and are faster than cache and RAM.
- ▶ They hold instructions, addresses, data, and intermediate results temporarily while work is in progress.
- ▶ Their capacity is very small, so they are used only for the values needed right now.

Speed order

Registers are fastest, then cache, then main memory (RAM), then secondary storage.

Key Registers and Their Purposes

Register	Holds
Program Counter (PC / IP)	Address of the next instruction to fetch
Instruction Register (IR)	Instruction currently being decoded/executed
Memory Address Register (MAR)	Address of the memory location to access
Memory Data Register (MDR)	Data being transferred to or from memory
Accumulator (ACC)	Intermediate arithmetic/logic results
General-purpose registers	Temporary operands and results
Status / flags register	Condition flags (zero, carry, sign, overflow)

PC/IP = Program Counter or Instruction Pointer.

The Program Counter: A Common Trap

- ▶ The **Program Counter (PC)** holds the address of the **next** instruction to be fetched.
- ▶ It does **not** hold the address of the instruction currently being executed.
- ▶ The **Instruction Register (IR)** holds the instruction that is currently being decoded or executed.

Trap alert

Near-neighbour confusion: PC = address of the **next** instruction; IR = the **current** instruction itself.

Clock Speed, Cores, and Threads

- ▶ **Clock speed** is measured in hertz (Hz); 1 GHz is one billion cycles per second.
- ▶ A **clock cycle** is one tick of the system clock; different instructions may take different numbers of cycles.
- ▶ A **core** is an independent processing unit inside the CPU; a quad-core chip has four cores.
- ▶ A **thread** is a stream of instructions; multiple threads can share or run on the available cores.

Office desktop example

A quad-core 2.4 GHz machine has four cores, each running at about 2.4 billion clock cycles per second.

Lesson 05: Recall

- ▶ The CPU has the ALU, the Control Unit, and the registers.
- ▶ The ALU performs arithmetic and logic on binary data using logic gates; the Control Unit only directs and issues control signals.
- ▶ Registers are the fastest, smallest storage, inside the CPU.
- ▶ PC holds the address of the **next** instruction; IR holds the **current** instruction; MAR holds an address; MDR holds data; the accumulator holds intermediate results.
- ▶ Clock speed is in Hz/GHz; cores and threads describe parallel execution.